Context-Free Session Types with Borrowing: Supplementary Material

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The organization of the supplement follows the structure of the main document. Topics are covered in the same sequence, so the two documents can be read side-by-side.

S-A Motivation — Remarks

The reader may ask why **select** still adheres to resource-passing style. The answer is simplicity of our system. While it is possible to define **select** with a borrowing-based interface, supporting it requires subtyping of session types. While subtyping is fine in theory, it turns out that it is undecidable for CFST [Padovani 2019]. Hence, it cannot be the basis for a complete type checker.

For borrowing from regular session types [Saffrich et al. 2025], a borrowing-style **select** operation is possible because subtyping for regular session types is decidable.

A borrowing-style **branch** operator (external choice) cannot be supported in CSTB. This operator selects between different continuations at run time and the residual channel depends on this run-time choice.

All CSTB examples are available as part of our supplement. Compared to the presentation in the main text, the split operators carry type parameters in the actual code to simplify type checking.

S-B Types: Session-Type Laws and Well Formedness Predicates

$$\textsc{U-Unit}\ \ \mathsf{unr}\ \mathtt{Unit} \qquad \textsc{U-Arr}\ \ \mathsf{unr}\ (T_1 \to^{\mathrm{M}}_{\omega,\epsilon} T_2) \qquad \textsc{U-Prod}\ \ \frac{\mathsf{unr}\ T_1 \qquad \mathsf{unr}\ T_2}{\mathsf{unr}\ (T_1 \otimes_d T_2)} \qquad \textsc{U-Variant}\ \ \frac{\forall \ell.\ \mathsf{unr}\ T_\ell}{\mathsf{unr}\ \langle \ell : T_\ell \rangle_{\ell \in L}}$$

$$\textsc{U-CtxEmpty}\ \ \mathsf{unr}\ \emptyset \qquad \textsc{U-CtxVar}\ \ \frac{\mathsf{unr}\ T}{\mathsf{unr}\ x : T} \qquad \textsc{U-CtxSeq}\ \ \frac{\mathsf{unr}\ \Gamma_1 \qquad \mathsf{unr}\ \Gamma_2}{\mathsf{unr}\ (\Gamma_1, \Gamma_2)} \qquad \textsc{U-CtxPar}\ \ \frac{\mathsf{unr}\ \Gamma_1 \qquad \mathsf{unr}\ \Gamma_2}{\mathsf{unr}\ (\Gamma_1 \parallel \Gamma_2)}$$

$$\textsc{TN-Skip}\ \ \mathsf{new}\ \mathtt{Skip} \qquad \textsc{TN-Send}\ \ \mathsf{new}\ !T \qquad \textsc{TN-Recv}\ \ \mathsf{new}\ ?T \qquad \textsc{TN-Var}\ \ \mathsf{new}\ X$$

$$\textsc{TN-Seq}\ \ \frac{\mathsf{new}\ S_1 \qquad \mathsf{new}\ S_2}{\mathsf{new}\ (S_1\ ;\ S_2)} \qquad \textsc{TN-Branch}\ \ \frac{\forall \ell.\ \mathsf{new}\ S_\ell}{\mathsf{new}\ \&\{\ell : S_\ell\}_{\ell \in L}} \qquad \textsc{TN-Choice}\ \ \frac{\forall \ell.\ \mathsf{new}\ S_\ell}{\mathsf{new}\ \oplus\{\ell : S_\ell\}_{\ell \in L}} \qquad \textsc{TN-Rec}\ \ \frac{\mathsf{new}\ S}{\mathsf{new}\ \mu X.S}$$

Fig. S-1. Unrestricted types and contexts ($\mathsf{unr}\ T$, $\mathsf{unr}\ \Gamma$); session types valid for `new` ($\mathsf{new}\ S$)

Laws for context-free session types.

$$\mathtt{Skip}\ ;\ S = S \qquad S\ ;\ \mathtt{Skip} = S \qquad S_1\ ;\ (S_2\ ;\ S_3) = (S_1\ ;\ S_2)\ ;\ S_3 \qquad \mu X.S = S[\mu X.S/X]$$

$$(\oplus\{\ell : S_\ell\}_{\ell \in L})\ ;\ S = \oplus\{\ell : S_\ell\ ;\ S\}_{\ell \in L} \qquad (\&\{\ell : S_\ell\}_{\ell \in L})\ ;\ S = \&\{\ell : S_\ell\ ;\ S\}_{\ell \in L}$$

Laws for duality $S^\perp$ (there is no dual for `Drop` and `Acq`).

$$\mathtt{Skip}^\perp = \mathtt{Skip} \qquad (S_1\ ;\ S_2)^\perp = {S_1}^\perp\ ;\ {S_2}^\perp \qquad !T^\perp = ?T \qquad ?T^\perp = !T$$

$$(\oplus\{\ell : S_\ell\}_{\ell \in L})^\perp = \&\{\ell : {S_\ell}^\perp\}_{\ell \in L} \qquad (\&\{\ell : S_\ell\}_{\ell \in L})^\perp = \oplus\{\ell : {S_\ell}^\perp\}_{\ell \in L} \qquad \mathtt{Close}^\perp = \mathtt{Wait}$$

$$\mathtt{Wait}^\perp = \mathtt{Close} \qquad (\mu X.S)^\perp = S[\mu X.S/X]^\perp$$

Fig. S-2. Session-type equality and duality

S-C Dynamics: Congruence and Reduction Rules

$$
\begin{array}{rcll}
u, v & ::= & c \mid x \mid \lambda x.e \mid \mathtt{inj}_i\, v \mid u \otimes v & \text{(Values)} \\
\mathcal{E} & ::= & \Box \mid (\mathcal{E} \mathbin{@_{\mathrm{L}}} e) \mid (v \mathbin{@_{\mathrm{L}}} \mathcal{E}) \mid (e \mathbin{@_{\mathrm{R}}} \mathcal{E}) \mid (\mathcal{E} \mathbin{@_{\mathrm{R}}} v) \mid \mathcal{E}; e \mid \mathcal{E} \otimes e \mid v \otimes \mathcal{E} & \\
 & & \mid \mathtt{inj}_i\, \mathcal{E} \mid \mathtt{let}\ x \otimes x = \mathcal{E}\ \mathtt{in}\ e \mid \mathtt{case}\, \mathcal{E}\, \{\overline{\mathtt{inj}_i\, x \Rightarrow e}\} & \text{(Evaluation contexts)} \\
\mathcal{F} & ::= & \langle \mathcal{E} \rangle & \text{(Process evaluation contexts)} \\
\mathcal{G} & ::= & \Box \mid \mathcal{G} \parallel P \mid (\nu[B][B])\mathcal{G} & \text{(Process contexts)}
\end{array}
$$

Fig. S-3. Values and evaluation contexts

E-App
$(\lambda x.e)\, v \longrightarrow_e e[v/x]$

E-Seq
$v; e \longrightarrow_e e$

E-PairElim
$\mathtt{let}\ x \otimes y = u \otimes v\ \mathtt{in}\ e \longrightarrow_e e[v/x][u/y]$

E-Unfold
$\mu f.e \longrightarrow_e e[\mu f.e/f]$

E-VariantElim
$$\frac{k \in L}{\mathtt{case}\,(\mathtt{inj}_k\, v)\,\{\mathtt{inj}_\ell\, x_\ell \Rightarrow e_\ell\}_{\ell \in L} \longrightarrow_e e_k[v/x_k]}$$

E-Ctx
$$\frac{e_1 \longrightarrow_e e_2}{\mathcal{E}[e_1] \longrightarrow_e \mathcal{E}[e_2]}$$

Fig. S-4. Expression reduction ($e \longrightarrow_e e$)

C-ParComm
$P_1 \parallel P_2 \equiv P_2 \parallel P_1$

C-ParAssoc
$P_1 \parallel (P_2 \parallel P_3) \equiv (P_1 \parallel P_2) \parallel P_3$

C-ParUnit
$\langle * \rangle \parallel P \equiv P$

C-ResSwap
$(\nu[B_1][B_2])P \equiv (\nu[B_2][B_1])P$

C-ResComm
$$\frac{(\mathsf{fv}(B_1) \cup \mathsf{fv}(B_2)) \cap (\mathsf{fv}(B_3) \cup \mathsf{fv}(B_4)) = \emptyset}{(\nu[B_1][B_2])(\nu[B_3][B_4])P \equiv (\nu[B_3][B_4])(\nu[B_1][B_2])P}$$

C-Extend
$$\frac{\mathsf{fv}(P_1) \cap (\mathsf{fv}(B_1) \cup \mathsf{fv}(B_2)) = \emptyset}{P_1 \parallel (\nu[B_1][B_2])P_2 \equiv (\nu[B_1][B_2])(P_1 \parallel P_2)}$$

Fig. S-5. Process congruence

S-D Algorithmic Typing

$$\Gamma_1 \overset{d}{\bowtie} \Gamma_2 = \begin{cases} \Gamma_1 \parallel \Gamma_2 & \text{if } d \in \{\mathbf{1}, \omega\} \\ \Gamma_1, \Gamma_2 & \text{if } d = \textsc{l} \\ \Gamma_2, \Gamma_1 & \text{if } d = \textsc{r} \end{cases} \qquad \Gamma|_X = \begin{cases} \Gamma_1|_X \parallel \Gamma_2|_X & \text{if } \Gamma = \Gamma_1 \parallel \Gamma_2 \\ \Gamma_1|_X, \Gamma_2|_X & \text{if } \Gamma = \Gamma_1, \Gamma_2 \\ x : T & \text{if } \Gamma = x : T \text{ and } x \in X \\ \emptyset & \text{otherwise} \end{cases}$$

Fig. S-6. Context join $\Gamma_1 \overset{d}{\bowtie} \Gamma_2$ joins two contexts according to the direction d, and context restriction $\Gamma|_X$ restricts a context to a set of variables X.

CS-Pair
$$T^* \otimes_d U^* \simeq T^{*\prime} \otimes_d U^{*\prime} \Longrightarrow \left\{T^* \simeq T^{*\prime}, U^* \simeq U^{*\prime}\right\}$$

CS-Sum
$$T^* + U^* \simeq T^{*\prime} + U^{*\prime} \Longrightarrow \left\{T^* \simeq T^{*\prime}, U^* \simeq U^{*\prime}\right\}$$

CS-Unit
$$\mathtt{Unit} \simeq \mathtt{Unit} \Longrightarrow \emptyset$$

CS-Arrow
$$T^* \to^m_{d,\epsilon} U^* \simeq T^{*\prime} \to^m_{d,\epsilon} U^{*\prime} \Longrightarrow \{T^{*\prime} \simeq T^*, U^* \simeq U^{*\prime}\}$$

CS-Swap
$$\frac{T^* \notin A}{T^* \simeq \alpha \Longrightarrow \{\alpha \simeq T^*\}}$$

CS-Context
$$\frac{T^* \simeq U^* \Longrightarrow \Delta'}{\Delta \cup \{T^* \simeq U^*\} \Longrightarrow \Delta \cup \Delta'}$$

Fig. S-7. Constraint simplification

S-E Implementation

S-E.1 Constraint Checking with FreeST

In the implementation, we use FreeST [Almeida et al. 2022, 2019, 2026] version 5.0 to perform equivalence checking. To this end, we generate little FreeST programs that typecheck if and only if the equivalence constraints hold.

For every equivalence constraint $T_1 \simeq T_2$, we generate the following FreeST program consisting of type definitions for T_1 and T_2 as well as an identity function mapping T_1 to T_2. For example, let $T_1 = \mu X.!\texttt{Int} ; !\texttt{String} ; X$ and $T_2 = \mu X.\texttt{Skip} ; !\texttt{Int} ; !\texttt{String} ; X$. For the constraint $T_1 \simeq T_2$ we generate the following FreeST program:

```
type X_T1 : 1S
type X_T1 = !Int ; !String ; X_T1

type T1 : 1S
type T1 = X_T1

type X_T2 : 1S
type X_T2 = Skip ; !Int ; !String ; X_T2

type T2 : 1S
type T2 = X_T2

id : T1 -> T2
id x = x
```

FreeST 5.0 requires specifying kinds for type definitions. Kind 1S says that the type needs to be a session type. If *T1* and *T2* are the translated types from the equivalence constraint, it is easy to see that this program only typechecks, if *T1* and *T2* are equivalent session types.

The main parts of the translation are:

- Translating inline recursion of the form $\mu X.S$ to explicit recursion.
- Translating mobility and direction annotations on function and pair types.
- Translating effects.

To translate a μ-type to explicit recursion, we generate a fresh name and associate the name with the recursion unrolled and translation applied. We return the associated name, instead of the type. We encode mobility, direction, and effects in a branch type with labels indicating the state of the annotation. Thus, the translation function $[\![T]\!]$, defined in fig. S-8, takes a type T and returns a pair $\langle F, U \rangle$ consisting of a FreeST type F and a list of explicit type definitions U.

S-E.2 Soundness and the Completeness of the translation

Definition S-E.1. The equivalence relation $\simeq$ on CSTB types is the least congruence closed under the equivalence rules from Thiemann and Vasconcelos [2016] extended with rules from Figure S-9.

Let F be a FreeST session type and U be a mapping from variables to FreeST session types. In the following, we write $F[U]$ for the pair of F and U where $\text{freevars}(F) \subseteq \text{dom}(U)$. This notation includes unfolding of definitions in the following way:

$$\frac{(X \mapsto F) \in U}{X[U] \simeq F[U]}$$

Lemma S-E.2 (Soundness of the translation). *If $T_1 \simeq T_2$, then $F_1[U_1] \simeq F_2[U_2]$ where $\langle F_1, U_1 \rangle = [\![T_1]\!]$ and $\langle F_2, U_2 \rangle = [\![T_2]\!]$.*

$$
\begin{aligned}
[\![\texttt{Unit}]\!] &= \langle (), \emptyset \rangle \\
[\![\texttt{Skip}]\!] &= \langle \texttt{Skip}, [\,] \rangle \\
[\![\texttt{Close}]\!] &= \langle \texttt{Close}, [\,] \rangle \\
[\![\texttt{Wait}]\!] &= \langle \texttt{Wait}, [\,] \rangle \\
[\![T_1 \to^{m}_{d,\epsilon} T_2]\!] &= \langle (F_1 \to F_2) \otimes_d \&\{\ell : \texttt{Skip}\}_{\ell \in \text{dir} \cup \text{mob} \cup \text{eff}}, U_1 \cup U_2 \rangle \quad \text{where} \\
&\quad \langle F_1, U_1 \rangle = [\![T_1]\!] \\
&\quad \langle F_2, U_2 \rangle = [\![T_2]\!] \\
&\quad \text{dir} = \begin{cases} \{\text{left}\} & \text{if } d = \text{L} \\ \{\text{right}\} & \text{if } d = \text{R} \\ \{\text{lin}\} & \text{if } d = \mathbf{1} \\ \emptyset & \text{if } d = \omega \end{cases} \\
&\quad \text{mob} = \begin{cases} \{\text{mobile}\} & \text{if } m = \text{M} \\ \{\text{static}\} & \text{if } m = \text{S} \end{cases} \\
&\quad \text{eff} = \begin{cases} \{\text{impure}\} & \text{if } \epsilon = \text{I} \\ \{\} & \text{if } \epsilon = \text{I} \end{cases} \\
[\![T_1 \otimes_d T_2]\!] &= \langle (F_1 \times F_2) \otimes_d \&\{\ell : \texttt{Skip}\}_{\ell \in \text{dir}}, U_1 \cup U_2 \rangle \quad \text{where} \\
&\quad \langle F_1, U_1 \rangle = [\![T_1]\!] \\
&\quad \langle F_2, U_2 \rangle = [\![T_2]\!] \\
&\quad \text{dir} = \begin{cases} \{\text{left}\} & \text{if } d = \text{L} \\ \{\text{right}\} & \text{if } d = \text{R} \\ \{\text{lin}\} & \text{if } d = \mathbf{1} \\ \emptyset & \text{if } d = \omega \end{cases} \\
[\![\langle \ell : T_\ell \rangle_{\ell \in L}]\!] &= \langle \langle \ell : F_\ell \rangle_{\ell \in L}, \bigcup_{\ell \in L} U_\ell \rangle \quad \text{where} \\
&\quad \langle F_\ell, U_\ell \rangle = [\![T_\ell]\!] \quad \text{for every } \ell \in L \\
[\![!T]\!] &= \langle !F, U \rangle \quad \text{where} \\
&\quad \langle F, U \rangle = [\![T]\!] \\
[\![?T]\!] &= \langle ?F, U \rangle \quad \text{where} \\
&\quad \langle F, U \rangle = [\![T]\!] \\
[\![S_1; S_2]\!] &= \langle F_1; F_2, U_1 \cup U_2 \rangle \quad \text{where} \\
&\quad \langle F_1, U_1 \rangle = [\![S_1]\!] \\
&\quad \langle F_2, U_2 \rangle = [\![S_2]\!] \\
[\![\oplus\{\ell : S_\ell\}_{\ell \in L}]\!] &= \langle +\{\ell : F_\ell\}_{\ell \in L}, \bigcup_{\ell \in L} U_\ell \rangle \quad \text{where} \\
&\quad \langle F_\ell, U_\ell \rangle = [\![S_\ell]\!] \quad \text{for every } \ell \in L \\
[\![\&\{\ell : S_\ell\}_{\ell \in L}]\!] &= \langle \&\{\ell : F_\ell\}_{\ell \in L}, \bigcup_{\ell \in L} U_\ell \rangle \quad \text{where} \\
&\quad \langle F_\ell, U_\ell \rangle = [\![S_\ell]\!] \quad \text{for every } \ell \in L \\
[\![\texttt{Drop}]\!] &= \langle !\texttt{Drop}, \emptyset \rangle \\
[\![\texttt{Acq}]\!] &= \langle ?\texttt{Drop}, \emptyset \rangle \\
[\![\mu X.S]\!] &= \langle X', U \cup \{X' \mapsto U\} \rangle \quad \text{where } X' \text{ is fresh and} \\
&\quad \langle F, U \rangle = [\![S\,[X'/X]]\!] \\
[\![X]\!] &= \langle X, \emptyset \rangle
\end{aligned}
$$

Fig. S-8. Translation of types to FreeST types.

Proof. We sketch a proof. The soundness proof requires two things: the translation of annotations in the arrow and product types preserves equivalence and the equivalence of the explicit recursion with the μ-binder syntax. For the former we can perform induction on the equivalence of $T_1 \simeq T_2$ and use equivalence and bisimulation lemmas from Thiemann and Vasconcelos [2016] for

$$\frac{T_1 \simeq T_1' \qquad T_2 \simeq T_2'}{T_1 \to^{m}_{d,\epsilon} T_2 \simeq T_1' \to^{m}_{d,\epsilon} T_2'} \qquad \frac{T_1 \simeq T_1' \qquad T_2 \simeq T_2'}{T_1 \otimes_d T_2 \simeq T_1' \otimes_d T_2'} \qquad \frac{T_1 \simeq T_1' \qquad T_2 \simeq T_2'}{T_1 + T_2 \simeq T_1' + T_2'}$$

Fig. S-9. Modified rules for $\simeq$

regular type and session type cases, respectively. The pairs we introduce for annotations can be handled trivially by the type equivalence rules. To show explicit recursion translation preserves equivalence, we can use the laws for μ-types from Thiemann and Vasconcelos [2016]. □

Lemma S-E.3 (Completeness of the translation). *Let* $\langle F_1, U_1 \rangle = [\![T_1]\!]$ *and* $\langle F_2, U_2 \rangle = [\![T_2]\!]$. *If* $F_1[U_1] \simeq F_2[U_2]$, *then* $T_1 \simeq T_2$.

Proof. Similar to S-E.2 but the opposite direction. □

S-F Metatheory

$$\textsc{B-ExpValueBlocked}\quad \text{Blocked } \langle * \rangle$$

$$\textsc{B-ExpConstBlocked}\quad \frac{c \notin \{\mathtt{new}, \mathtt{fork}, \mathtt{lsplit}, \mathtt{rsplit}, \mathtt{drop}, \mathtt{discard}\}}{\text{Blocked } \mathcal{F}[c\ v]}$$

$$\textsc{B-ParBlocked}\quad \frac{\text{Blocked } P_1 \qquad \text{Blocked } P_2}{\text{Blocked } P_1 \parallel P_2}$$

$$\textsc{B-NuBlocked}\quad \frac{\{x, y\} \cap \mathsf{BC}(P) \neq \{x, y\} \qquad \text{Blocked } P}{\text{Blocked } (\nu[xB_1][yB_2])P}$$

$$\textsc{B-NuBlockedAcqLeft}\quad \frac{x \notin \mathsf{AC}(P) \qquad \text{Blocked } P}{\text{Blocked } (\nu[\|xB_1][yB_2])P}$$

$$\textsc{B-NuBlockedAcqRight}\quad \frac{y \notin \mathsf{AC}(P) \qquad \text{Blocked } P}{\text{Blocked } (\nu[xB_1][\|yB_2])P}$$

$$\textsc{B-NuBlockedAcqBoth}\quad \frac{x \notin \mathsf{AC}(P) \qquad y \notin \mathsf{AC}(P) \qquad \text{Blocked } P}{\text{Blocked } (\nu[\|xB_1][\|yB_2])P}$$

where

$$\begin{array}{ll}
\mathsf{BC}(\mathcal{F}[\mathtt{send}\,(v \otimes x)]) = \{x\} & \mathsf{BC}(\mathcal{F}[\mathtt{recv}\,x]) = \{x\} \\
\mathsf{BC}(\mathcal{F}[\mathtt{select}\,k\,x]) = \{x\} & \mathsf{BC}(\mathcal{F}[\mathtt{branch}\,x]) = \{x\} \\
\mathsf{BC}(\mathcal{F}[\mathtt{close}\,x]) = \{x\} & \mathsf{BC}(\mathcal{F}[\mathtt{wait}\,x]) = \{x\} \\
\mathsf{BC}(\mathcal{F}[\mathtt{acquire}\,x]) = \emptyset & \mathsf{BC}(\mathcal{F}[\mathtt{drop}\,x]) = \emptyset \\
\mathsf{BC}(\mathcal{F}[\mathtt{lsplit}\,x]) = \emptyset & \mathsf{BC}(\mathcal{F}[\mathtt{rsplit}\,x]) = \emptyset \\
\mathsf{BC}(\mathcal{F}[\mathtt{fork}\,v]) = \emptyset & \mathsf{BC}(\mathcal{F}[\mathtt{new}\,e]) = \emptyset \\
\mathsf{BC}(\mathcal{F}[\mathtt{discard}\,x]) = \emptyset &
\end{array}$$

$$\begin{array}{ll}
\mathsf{AC}(\mathcal{F}[\mathtt{send}\,(v \otimes x)]) = \emptyset & \mathsf{AC}(\mathcal{F}[\mathtt{recv}\,x]) = \emptyset \\
\mathsf{AC}(\mathcal{F}[\mathtt{select}\,k\,x]) = \emptyset & \mathsf{AC}(\mathcal{F}[\mathtt{branch}\,x]) = \emptyset \\
\mathsf{AC}(\mathcal{F}[\mathtt{close}\,x]) = \emptyset & \mathsf{AC}(\mathcal{F}[\mathtt{wait}\,x]) = \emptyset \\
\mathsf{AC}(\mathcal{F}[\mathtt{acquire}\,x]) = \{x\} & \mathsf{AC}(\mathcal{F}[\mathtt{drop}\,x]) = \emptyset \\
\mathsf{AC}(\mathcal{F}[\mathtt{lsplit}\,x]) = \emptyset & \mathsf{AC}(\mathcal{F}[\mathtt{rsplit}\,x]) = \emptyset \\
\mathsf{AC}(\mathcal{F}[\mathtt{fork}\,v]) = \emptyset & \mathsf{AC}(\mathcal{F}[\mathtt{new}\,e]) = \emptyset \\
\mathsf{AC}(\mathcal{F}[\mathtt{discard}\,x]) = \emptyset &
\end{array}$$

Fig. S-10. Blocked Processes (Blocked P)

S-F.1 Type Safety for Declarative Typing

Process preservation and progress are proved in Agda, see **??**. The earlier paper proof of process progress for the imprecise blocked predicate is retained below as a comment.

S-F.2 Soundness and Completeness for Algorithmic Typing

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